

Wanglong Lu

Senior Data Scientist | Applied Multimodal Intelligence

St. John's, NL, Canada | lwlxhl@gmail.com | Website | GitHub | LinkedIn | Google Scholar

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Research & Professional Profile

Computer Science Ph.D.; **30 scholarly works overall** (author-reported), including **six published first-author papers**, co-first-authored DocPure and corresponding-authored EchoSR. Senior Data Scientist at Nasdaq and applied multimodal intelligence researcher, focused on generative vision, efficient adaptation, financial ML and cloud-based ML systems.

Professional Experience

Nasdaq

St. John's, NL, Canada

Senior Data Scientist, AI/Analytics

Mar. 2025 – Present

- Delivered **MLP application deployment and integration** into a **SageMaker workspace** from an existing cross-team UI/backend prototype; configured networking/IAM, **PostgreSQL state** and **FastAPI/Mangum on Lambda**. The platform is at the **internal deployment stage**.
- Designed cross-scale ensemble feature selection that removed **77.2% of input features** while retaining competitive performance on CFML data; presented the algorithm to more than 60 attendees.
- Built **SageMaker Job Submitter for MLP**: script/dependency packaging for **Processing, Training, HPO and Pipelines**; reduced **new-task configuration and submission preparation from 7 days to 1 day (about 86%)**, not training runtime.
- Designed **feature engineering with PyTorch/CUDA-accelerated feature-label AUROC screening**; used **Amazon Bedrock** to assess candidate features' semantic plausibility.
- Developed **PagerDuty RAG assistant v2** on the company's GenAI platform; **now used internally and recognized in a company shout-out**. Retrieves incident context and troubleshooting guidance for on-call engineers.
- Mentored **cheque OCR** research on perceptually aligned quality metrics, culminating in a student presentation; guided **tabular-data generation** leading to a **StDDPM manuscript submission**.
- Mentored **Seeam (work-term student)** on an **input payload checker** from Jun. 2026, and **Daniel (Mitacs student)** on **GNN-based crime-ring detection** from Sep. 2026.

AI Algorithm Intern (two placements)

May 2024 – Jan. 2025

- Proposed **TextDoctor**: structure-pyramid prediction and patch-pyramid diffusion for document/cheque inpainting; **8K inpainting on a 24GB RTX 3090**. Python/PyTorch, AWS/SLURM/CUDA. **First-author manuscript submitted to Engineering Applications of Artificial Intelligence (EAAI)**.
- Evaluated **7 document datasets** without dataset-specific fine-tuning. FUNSD, TextDoctor (GSDM): **55.38% PaddleOCR word accuracy vs. 45.24% DocDiff (+10.14 percentage points)**.
- Presented research after **both internship placements**: earlier talk about **40–60 attendees**; final talk **39 cross-functional attendees**.

AI Intern – Beaufort Solutions

Jul. 2022 – Nov. 2022

St. John's, NL, Canada

- Led method design of **TEG**, a CLIP-based **few-shot adaptation** approach for personalized photo-book curation; introduced a text-embedding-guided classifier and auxiliary classification loss to improve learning from limited labels.
- Led Theme25 collection pipeline (**35,655 annotated images, 25 themes**); evaluated Theme25, CIFAR100, and ImageNet on AWS/SLURM; published in **Journal of Electronic Imaging (2024)**.

AI Algorithm Intern – Zhongkong Hanlian

Dec. 2017 – Aug. 2018

Hangzhou, China

- Designed **ResNet–DenseNet fusion** for vehicle-logo recognition; built Java annotation tools and a dataset for training and evaluation; **deployed with Caffe for real-time inference at community security gates**.

Education

Ph.D., Computer Science – Memorial University of Newfoundland, Canada	2021–2025
M.Eng., Computer Software and Theory – Wenzhou University, China	2018–2021
B.Eng., Digital Media Technology – Communication University of Zhejiang, China	2014–2018

Selected Research & Engineering Projects

UltraDiffEdit – High-Resolution Generative Editing [Code] TNNLS 2026; first author

- Developed multi-patch encoding, global-local consistency denoising, and progressive sampling to extend pretrained latent diffusion models to real-image editing without fine-tuning.
- Demonstrated **8K editing on one RTX 3090**; achieved **UHRSEdit (4K–8K) crop-FID 7.45 vs. 18.76 for SDXL+bicubic (60.3% lower)**. Video.
- Built **three benchmarks totaling 1,188 images**: DIV2KEdit, Syn2KEdit, and UHRSEdit; evaluated pretrained SDXL and ControlNet without fine-tuning.

FACEMUG – Multimodal Local Facial Editing TVCG 2025; first author

- Developed a unified editing framework for text, sketches, semantic maps, color maps, exemplars, and attribute labels, with multimodal feature fusion and self-supervised latent warping.
- Published in **TVCG 31(9), 5130–5145 (2025)**, following online publication in 2024.
- Achieved **FFHQ FID 23.16 vs. 52.39 for Unite&Conquer** under sketch/semantic conditioning; received **77.88% preference** in a **52-participant CelebA-HQ study** with sketch/semantic/text guidance. Publication.

AdaMSS – Parameter-Efficient Adaptation [Code] NeurIPS 2025; second author

- Co-developed adaptive multi-subspace updates; ViT-Large averaged **90.13% vs. LoRA's 85.40% (+4.73 percentage points)** across seven tasks, with **241K vs. 1.57M trainable parameters (15.4%)**.
- LLaMA 2-7B reached **50.80% GSM8K accuracy with 4M trainable parameters** vs. 49.05% with 6,738M for full fine-tuning. Method **integrated into Hugging Face PEFT**.

DocPure – Unified Document Restoration [Code] TCSVT 2026; co-first author

- Co-developed degradation-aware structure extraction and wavelet modulation; tested **one prompt-free model on 12 datasets** for deblurring, denoising, compression artifacts, and deshadowing.
- On TDD deblurring, achieved **32.55 dB PSNR vs. DocRes's 28.34 dB (+4.21 dB)** and **61.27% PaddleOCR word accuracy vs. 55.66% (+5.61 percentage points)**. Benchmarks.

GRIG – Data-Efficient Image Inpainting [Code] CVM 2025; first author

- Designed CNN–Transformer iterative residual refinement and forgery-patch adversarial training for data-efficient inpainting on **10 few-shot datasets with 18–750 training images**.
- Author-manuscript results: **3.70–26.48% lower FID vs. each dataset's next-best baseline at 50–60% masking**; benchmarks span medical, animal, cartoon, and art images. Published paper.

Visual Style Prompt Learning [Code] Pattern Recognition 2025; first author

- Developed latent-diffusion style prompts and style-modulated aggregation with **four denoising steps**; achieved **12.3 FPS at 512 × 512 on an RTX 3080 Ti**.
- Achieved **WebPhoto-Test FID 74.25 vs. CodeFormer's 83.17 (10.7% lower)**; best mean rank (3.67) across FID/NIQE on three real-world test sets. Paper.
- On **CelebA-Test**, reduced **FAN landmark NME from 6.08% to 2.43%** and raised **HSEmotion agreement from 79.73% to 86.03%**; reference landmarks and emotion labels are model predictions on clean images.

Local Agentic Code Editor Nasdaq internal prototype; 2026

Exploratory side project: API-driven code editing and local tool orchestration

- Built a working local AI code-editor prototype with **model APIs and agentic routing** across code generation, review and shell-tool execution.
- Implemented an iterative debugging loop that **diagnoses bugs, edits code, applies fixes and uses execution feedback** to determine the next action.

Peer-Reviewed Publications

Tuning-Free Latent Diffusion Models for Ultrahigh-Resolution Image Editing

Wanglong Lu, Lingming Su, Kaijie Shi, Minglun Gong, Xiaogang Jin, Hanli Zhao, Xianta Jiang; First author
IEEE Transactions on Neural Networks and Learning Systems, 1-15, 2026

UHDRes: Ultra-High-Definition Image Restoration via Dual-Domain Decoupled Spectral Modulation

Shihao Zhao, **Wanglong Lu**, Binhao Wang, Tao Wang, Kaihao Zhang, Hanli Zhao
IEEE Transactions on Circuits and Systems for Video Technology, 2026

DocPure: Prompt-Free Unified Document Restoration via Degradation-Aware Structure-Guided Wavelet Modulation

Lingming Su, **Wanglong Lu**, Tao Wang, Kaihao Zhang, Nan Zhang, Liyan An, Hanli Zhao; Co-first author
IEEE Transactions on Circuits and Systems for Video Technology, 2026

EchoSR: Efficient Context Harnessing for Lightweight Image Super-Resolution

Hanli Zhao, Binhao Wang, Shihao Zhao, Tao Wang, Kaihao Zhang, **Wanglong Lu**; Corresponding author (sixth author)
Information Fusion 135, 104471, 2026

FACEMUG: A Multimodal Generative and Fusion Framework for Local Facial Editing

Wanglong Lu, Jikai Wang, Xiaogang Jin, Xianta Jiang, and Hanli Zhao; First author
IEEE Transactions on Visualization and Computer Graphics 31(9), 5130-5145, 2025 (online 2024)

AdaMSS: Adaptive Multi-Subspace Approach for Parameter-Efficient Fine-Tuning

Jingjing Zheng, **Wanglong Lu**, Yiming Dong, Chaojie Ji, Yankai Cao, Zhouchen Lin; Second author
NeurIPS 2025

Visual style prompt learning using diffusion models for blind face restoration

Wanglong Lu, Jikai Wang, Tao Wang, Kaihao Zhang, Xianta Jiang, Hanli Zhao; First author
Pattern Recognition 161, 111312, 2025

Toward Biosignals-Free Autonomous Prosthetic Hand Control via Imitation Learning

Kaijie Shi, **Wanglong Lu**, Hanli Zhao, Vinicius Prado da Fonseca, Ting Zou, Xianta Jiang
IEEE Transactions on Neural Systems and Rehabilitation Engineering 33, 3544-3554, 2025

GRIG: Data-Efficient Generative Residual Image Inpainting

Wanglong Lu, Xianta Jiang, Xiaogang Jin, Yong-Liang Yang, Minglun Gong, Kaijie Shi, Tao Wang, Hanli Zhao; First author
Computational Visual Media 11(6), 1329-1361, 2025

Do Inpainting Yourself: Generative Facial Inpainting Guided by Exemplars

Wanglong Lu, Hanli Zhao, Xianta Jiang, Xiaogang Jin, Yongliang Yang, Min Wang, Jiankai Lyu, and Kaijie Shi; First author
Neurocomputing 617, 128996, 2025

Degradation-Aware Frequency-Separated Transformer for Blind Super-Resolution

Hanli Zhao, Binhao Wang, **Wanglong Lu**, Juncong Lin
International Conference on Computational Visual Media, 2025

Real-time dual-eye collaborative eyeblink detection with contrastive learning

Hanli Zhao, Yu Wang, **Wanglong Lu**, Zili Yi, Jun Liu, Minglun Gong
Pattern Recognition, 2025

Handling the Non-smooth Challenge in Tensor SVD: A Multi-objective Tensor Recovery Framework

Jingjing Zheng, **Wanglong Lu**, Wenzhe Wang, Yankai Cao, Xiaoqin Zhang, and Xianta Jiang
ECCV, 2024

Restoring Vision in Hazy Weather with Hierarchical Contrastive Learning

Tao Wang, Guangpin Tao, **Wanglong Lu**, Kaihao Zhang, Wenhan Luo, Xiaoqin Zhang, Tong Lu.
Pattern Recognition Volume 145, January 2024, 109956

TEG: image theme recognition using text-embedding-guided few-shot adaptation

Jikai Wang, **Wanglong Lu**, Yu Wang, Kaijie Shi, Xianta Jiang, Hanli Zhao

Journal of Electronic Imaging 33(1), 013028, 2024

Detecting Blinks from Wearable Cameras using Spatial-Temporal-Aware Deep Network Learning

Yu Wang, **Wanglong Lu**, Hanli Zhao, Xianta Jiang, Bin Zheng, M. Stella Atkins.

ETRA 23, 2023

Efficient Generative Face Completion with Perceptual Deblurring

Hanli Zhao, Ying Liu, **Wanglong Lu**, Xiaogang Jin, Hui Huang, Kaijie Shi

Journal of Computer-Aided Design & Computer Graphics 34(9), 1420-1431, 2022

Generative image inpainting with enhanced gated convolution and Transformers

Min Wang, **Wanglong Lu**, Jiankai Lyu, Kaijie Shi, Hanli Zhao.

Displays 76, 102321, 2022

Probability-Based Channel Pruning for Depthwise Separable Convolutional Networks

Han-Li Zhao, Kai-Jie Shi, Xiaogang Jin, Ming-Liang Xu, Hui Huang, **Wanglong Lu**, Ying Liu.

Journal of Computer Science and Technology 37, 584-600, 2022

Image Splicing Tamper Detection Based on Multi-Scale Feature Priors [in Chinese; translated title]

Jiankai Lyu, **Wanglong Lu**, et al.; Second author

China Sciencepaper 17(11), 2022

Category-consistent deep network learning for accurate vehicle logo recognition

Wanglong Lu, Hanli Zhao, Qi He, Hui Huang, Xiaogang Jin.; First author

Neurocomputing 463, 623-636, 2021

High-quality retinal vessel segmentation using generative adversarial network with a large receptive field

Hanli Zhao, Xiqing Qiu, **Wanglong Lu**, Hui Huang, Xiaogang Jin.

International Journal of Imaging Systems and Technology 30(3), 828-842, 2020

Preprints & Submitted Manuscripts

Simulation-Driven Imitation Learning for Biosignals-Free Shared-Autonomy Prosthetic Grasping

Kaijie Shi, **Wanglong Lu**, Han Chen, Vinicius Prado da Fonseca, Ting Zou, Hanli Zhao, Xianta Jiang

arXiv preprint arXiv:2606.07389, 2026

TextDoctor: Unified Document Image Inpainting via Patch Pyramid Diffusion Models

Wanglong Lu, Lingming Su, Jingjing Zheng, Vinícius Veloso de Melo, Farzaneh Shoeleh, John Hawkin, Terrence Tricco, Hanli Zhao, Xianta Jiang; First author

arXiv preprint, 2025; *submitted to Engineering Applications of Artificial Intelligence (EAAI)*

StDDPM: Sequential Financial Synthetic Data Generation using T-distribution based Diffusion Model

Omid Tarkhaneh, Hamideh Mehri, **Wanglong Lu**, Farzaneh Shoeleh, Vinicius Veloso de Melo

Research Square preprint, 2025; *submitted*

Promptrr: Diffusion models as prompt generators for single image reflection removal

Tao Wang, **Wanglong Lu**, Kaihao Zhang, Wenhan Luo, Tae-Kyun Kim, Tong Lu, Hongdong Li, Ming-Hsuan Yang

ArXiv, 2024

Dissertation

Generative Models for Semantic Facial Image Editing: Multimodal Approaches

Wanglong Lu

PhD thesis, Memorial University of Newfoundland, 2025

Technical Skills

Programming: Python, PyTorch, SQL, Git, Bash, C++, Java, CUDA, OpenCV.

ML & GenAI: Latent diffusion, GANs, Transformers, multimodal conditioning, vision-language models, PEFT/LoRA, Hugging Face, RAG, feature selection, experimental evaluation.

Infrastructure: FastAPI, Mangum, PostgreSQL; AWS SageMaker, Bedrock, Lambda, S3, Docker, SLURM.

Additional Engineering Project

AI Headshot Generation: Implemented model invocation and deployed the generation application.

Teaching, Mentorship & Talks

- Mentored **12 students** in generative vision, image restoration/recognition, and applied ML research.
- Cheque OCR mentorship identified promising image-quality metrics consistent with human perception and supported text-extraction evaluation; the student completed a research presentation.
- Guided sequential financial-data generation research leading to **StDDPM**, a submitted manuscript with a Research Square preprint; evaluated on **1.06 million Czech banking transactions across 4,500 accounts**.
- Teaching assistant at Memorial: Data Structures & Algorithms (2021); Introduction to Machine Learning (2024).
- Guest lecturer, Bio-Inspired Robotics (ENGI 9986), Memorial University, Feb. 2025: a 150-minute lecture on Deep Generative Models and Applications for 10 graduate students.
- Invited speaker, IEEE Newfoundland and Labrador Section AGM, Mar. 2026: Generative Models for Semantic Image Editing: Multimodal Approaches.
- Research talks for academic/industry audiences; 2026 feature-selection presentation to **60+ attendees**.

Academic Service

Invited reviewer, 2026: IEEE TPAMI, IEEE TMM, IEEE Journal of Biomedical and Health Informatics (JBHI), Pattern Recognition, Applied Soft Computing, Engineering Applications of Artificial Intelligence, Scientific Reports, Neurocomputing, Knowledge-Based Systems, and ECCV.

2025: IEEE TIP, IEEE TMM, Pattern Recognition, Neurocomputing, CAD/Graphics, Advances in Manufacturing, Signal, Image and Video Processing, IEEE SPL, Multimedia Systems, Journal of Machine Learning and Cybernetics.

2024: IEEE TMM, IEEE TCSVT, Knowledge-Based Systems, Journal of Medical Systems, Cluster Computing, Journal of Visual Communication and Image Representation, Displays, The Visual Computer, and ETRA PETMEI.

Granted Patents

Co-inventor on 5 granted Chinese invention patents. All grant certificates list Wenzhou University as the assignee; titles are translated.

- Probability-based channel pruning for MobileNetV1. ZL202110903135.9; granted 2023-07-25; inventor 4/5.
- Category-consistent deep learning for image recognition. ZL202110408724.X; granted 2023-06-30; inventor 2/4.
- CNN-based intelligent vehicle-logo detection. ZL202010139068.3; granted 2021-09-03; inventor 2/2.
- GAN-based intelligent retinal-vessel image segmentation. ZL201910884346.5; granted 2021-07-06; inventor 2/4.
- CNN deep-feature-based vehicle-logo recognition. ZL202010139043.3; granted 2021-05-11; inventor 2/3.

Selected Honors & Awards

- Computer Innovation and Entrepreneurship Award, Wenzhou Computer Federation; first prize, team leader (2024).
- Genesis entrepreneur training program, Memorial University (2024).
- Outstanding Research Award (Ph.D.), Computer Science, Memorial University (2023).
- Mitacs Accelerate Award (2022).
- Outstanding Graduate, Wenzhou University (2020); Outstanding Graduate of Zhejiang Province (2018).
- Chinese Graduate Mathematical Modeling Competition, third prize (2019); Lanqiao Cup C/C++ Programming Competition, second prize (2019); CCPC regional competition, third prize (2019).